

Technology Stack

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Name	AI Study Buddy API
Maximum Marks	2 Marks

Technology Stack:

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Node.js, Express.js
AI Technology	Artificial Intelligence, Natural Language Processing (NLP)
Database	MongoDB
Development Tool	Visual Studio Code
Version Control	GitHub

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML, CSS, JavaScript</i>	<i>Creates a simple, responsive, and user-friendly interface for students.</i>

2	Backend / Server-Side	<i>Node.js, Express.js</i>	<i>Handles user requests, processes queries, and connects the frontend with the AI module.</i>
3	Database / Data Storage	<i>MongoDB</i>	<i>Stores user information and application data efficiently.</i>
4	Cloud / Hosting / Deployment	<i>Localhost / Web Server</i>	<i>Used to run and test the web application during development.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Helps manage project code, track changes, and support team collaboration.</i>

6	Third-Party APIs / Other Tools	<i>AI / NLP API</i>	<i>Provides AI-based question answering and simple explanations for student queries.</i>
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